



Course Guide Book for Basic Electricity and Electronics

1. General Information of The Course

Course Name:	Basic Electricity and Electronics
Course Code:	ECEng 2202
Course Credits Points:	5 hrs.
Contact Hours:	2hrs lectures supported by 1hr tutorials and 2 lab per week
Course Module:	Electrical Engineering
Course Category:	Major
Status of Course:	Compulsory
Pre-Requisites:	None
Students' Department:	Mechanical Engineering
Degree Program:	BSC in Mechanical Engineering
Students Admission:	Regular
Year of Study:	2
Semester:	2 nd
Academic Year:	2024/2025
Course Duration:	From March 04, 2025, to April 03, 2025
Teaching & Learning Methods:	Lecture supported by tutorial, assignment, and project.
Attendance Requirement:	Minimum 85% during lecture and tutorial hours
Course Chair:	Nebiyu Y.
Course Instructor:	Estifanos A.
	Email: estifabeje@gmail.com
	Office: Ghion: 102
Consultation:	Monday afternoon 2:00 PM-3:00 PM
	Thursday morning 09:00AM -10:00AM

2. Course Description

Student will come to know about electrical circuit component and analysis methods, electromagnetic induction theory, A.C fundamentals and load characteristics, Semiconductor theory, P-N junction diode, transistor configuration, performance & characteristics.

3. Objectives Of the Course

The main objective of the coursework is to present the students to know about electrical circuit component and analysis methods, A.C fundamentals and load characteristics.

In particular, the specific objectives of the course are:

1. To provide a good conceptual grounding in to the basic principles of electricity and electronics

2. To know how circuit is analyzed
3. To understand the application of different electronics components & its characteristics
4. To know Theory of semiconductor devices and their application area

4. Course Learning Outcomes (CLO)

At the end of the course, students will be able to:

No	Course Learning Outcome (CLO)	Performance Indicators (PIs)	Program Student Outcome (PSO)
1	Demonstrates knowledge of the basic principles of electricity and electronics.	1-2	1
2	Carries out circuit analysis to determine equivalent resistance, voltage and power flow over using different types of methods.	1-2	1,2
3	Applies various theorems toward solving circuit analysis.	1-3	2,3
4	Able to use color coding to determine the resistance values in electrical devices.	3-4	1,2,3
5	Applies theory of semiconductor devices in their application area to understand the real applications	5-7	3,4

5. The Course Time Schedule and Teaching Methodology

The following topics will be covered in this course within this semester.

Table 1. Time Schedule and Teaching Methodology of the course

Week	Course Contents	Teaching Methodology and References	Attaining student outcome
[1,2]	Chapter 1: Basics of Electricity 1.1 Charge, Voltage and Current 1.2 Energy and Power 1.3 Coulomb's law and electric field 1.4 Magnetic field and Faraday's Law 1.5 Electric sources	✓ At least two examples should be done. ✓ Lecture, reading, class work, tutorials Textbook 1 p4-25 Ref. 1 p277-339	Outcome 1, & 2
Mid			
[3,4]	Chapter 2: Electrical Circuit Analysis and Theorems 2.1 Kirchhoff's laws (KVL & KCL); 2.2 Circuits simplifications (series and parallel connections of R, L, C and sources); 2.3 Star (Y) - delta (Δ) transformation; 2.4 Mesh analysis; nodal analysis; Steady state AC Circuit, 2.5 Average and Effective (RMS) Values 2.6 Phasor Representations and analysis of sinusoids, 2.7 Linearity and the superposition theorem;	✓ At least four examples should be done. ✓ Lecture, reading, class work, tutorials Textbook 1 p28-142; Textbook 1 p393-413;	Outcome 2, & 3

	2.8 Thevenin's and Norton's theorems; maximum power transfer		
[4,6]	Chapter 3: Basic Electronics 3.1 Semiconductor materials and their types 3.2 Semiconductor diodes and their applications 3.3 Bipolar junction transistors 3.4 Field effect transistors	✓ At least two examples should be done. ✓ Lecture, reading, class work, tutorials Textbook 1 p165-181; Ref. 1 p450-502;	Outcome 4
Final Exam			

6. Assessment and Evaluation

The course utilizes a variety of continuous assessment and evaluation methods to ensure the successful achievement of course learning outcomes. These methods include assignments, lab reports, quizzes, midterm exams, final exams, classwork, and presentations. All assignments and tasks must be completed within the specified deadlines. If you encounter any difficulties with the assignments or tests, you are free to contact me for assistance.

Home works, Assignments, and lab reports: Assignments may involve collaborative work for analyzing and developing general approaches. However, unless specified otherwise, each individual must complete their submissions. Assignments and lab reports may entail both individual and group components, with equal participation expected from all group members. While collaboration is encouraged, submitting another individual's work as one's own constitutes academic dishonesty and will incur penalties. All group members must contribute equally. Timely submission is mandatory, as late submissions will not be graded. Oral exams will accompany assignments and reports, requiring thorough preparation from all members. Additionally, the instructor reserves the right to assign additional tasks or assignments as necessary.

Quizzes: Quizzes serve to assess students' current understanding of the coursework. They are conducted in class following lectures or tutorial sessions.

Mid-term Exam and Final Exam: The exams will be closed book. During an exam, all necessary materials will be provided by the examiner.

Table 2. Summary of Assessments

No	Assessment type	Mark allotted	Content Coverage	Attaining course learning outcome (CLO)	Time Schedule
1	Assignment 1	10%	Chapter 1	1	Week 1
2	Quiz Exam 1	5%	Chapter 1	1	Week 2
3	Mid exam	25%	Chapter 1	1,2	Week 3
4	Lab report	5%	Chapter 2 & 3	1,2,3,4	Week 3
7	Lab Exam	5%	Chapters 1&2	2,3,4	Week 4,5
8	Final Exam	50%	Chapters 2 & 3	1,2,3,4,5	Week 6
	Total	100%			

7. Laboratory session

Table 3. Summary of laboratory session content and required material

No	Schedule	Laboratory Topic	Short Description of Laboratory	Attaining course outcome
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1	Week 2	Resistance Color code	Determining the value of resistors using their color code.	1
2	Week 3	Series and parallel resistance Measurement	Evaluating the measured and calculated equivalent resistance value in series, parallel and series-parallel combinations.	1 & 3
3	Week 4	Ohms Law	Verifying the relation between voltage, resistance and current.	2 & 3
4	Week 5	Verification of various Theorems	Concluding various theorems in electrical circuits using their measured values.	3

8. Course Policy

Throughout this course, all students are expected to follow the student code of behavior as outlined in the Bahir Dar University Senate Legislation, December 2020. Academic misconduct, such as cheating, fabrication, and plagiarism, will not be permitted and will be quickly reported to the proper authorities for action.

Students who miss more than 15% of the semester's classes will not be eligible to take the final exam. Any student who surpasses the 15% absence level will receive a grade of 'NG' (No Grade) and must present valid reasons for their absence, according to Senate law, Art. 174 for NG management.

All students are required to have complete records of course assessments and evaluations. If a student misses a quiz, midterm exam, or any other continuous assessment without a valid reason, they will receive a score of zero. However, if a student fails to attend the final exam or completes a term paper/project without a valid reason, the instructor will record an "NG" according to Senate legislation, Art. 171.

9. Grading system

The grading scale of the course is fixed scale according to senate legislation of Art. 177.

10. Text Books

1. C.K. Alexander & M. N.O. Sadiku, Fundamentals of Electric Circuits, 5th Edition, NY, 2011

11. Reference Books

1. Boylstad : Engineering Circuit Analysis, McGraw-Hill Higher Education : Jan, 2006
2. Charles A. Schuler, Electronics: Principles and Applications, Jan 18, 2007
3. P.S. Bimbhra, Generalized Theory of Electrical Machines, 2002.

12. Approval

The following bodies confirmed the course guidebook accordingly.

Position	Name	Signature:	Date
Course Instructors	<u>Estifanos A.</u>	_____	<u>26/2/25 G.C</u>
Course Chair	<u>Nebiyu Y.</u>	_____	<u>26/2/25 G.C</u>
Chair Holder	<u>Alebachew K.</u>	_____	<u>26/2/25 G.C</u>
V/ Dean	<u>Gashaye L.</u>	_____	<u>26/2/25 G.C</u>